

JABstem

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Past Papers

Nat 5

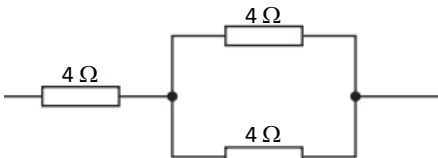
Physics

2015 Marking Scheme

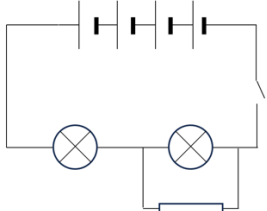
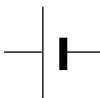
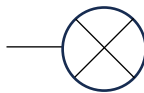
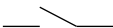
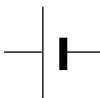
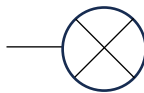
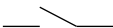
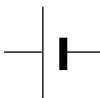
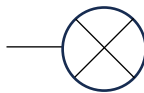
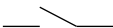
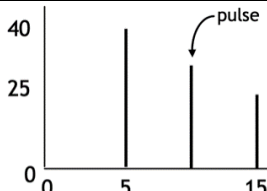
Grade Awarded	Mark Required (/100)	% candidates achieving grade
A	68+	28.9%
B	58+	23.3%
C	48+	21.8%
D	43+	8.5%
No award	<43	17.4%

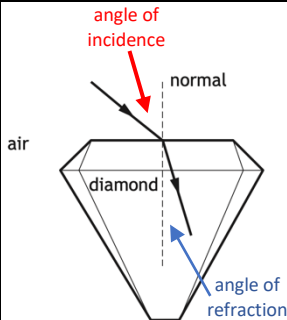
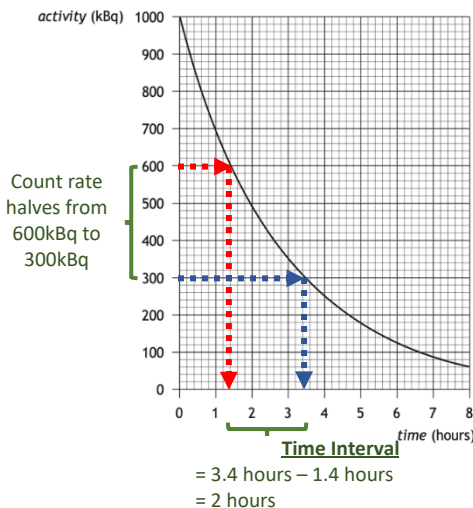
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	13.0 /20	32.1 /60	12.7 /20

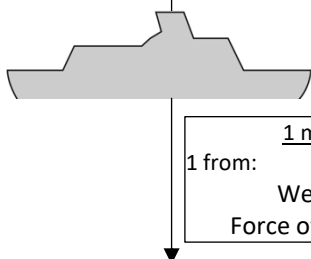
2015 Nat5 Physics Marking Scheme

Question	Answer	Physics Covered																																			
1	A	X		Y			Z																														
		Ohmmeter measuring resistance		Voltmeter measuring voltage			Ammeter measuring current																														
		Ohmmeter is placed in series with component. No cell required.		Voltmeters are place in parallel around the component being measured			Ammeters are connected in series in a circuit.																														
2	A	Statement I - Correct		Statement II - Incorrect			Statement III - Incorrect																														
		Voltage is defined as a measure of the energy given to the charge carriers in a circuit		A cell or battery has a voltage at all times whether connected in a circuit or unconnected.			Electrons are free to move in a <i>conductor</i>																														
3	C	Combining Parallel Resistors:				Combining Series Resistors:																															
																																					
		$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_T} = \frac{1}{4} + \frac{1}{4}$ $\frac{1}{R_T} = \frac{2}{4}$ $R_T = 2 \Omega$				$R_T = R_1 + R_2 = 4\Omega + 2\Omega = 6 \Omega$																															
4	E	$P = \frac{V^2}{R} \therefore 920 = \frac{(230)^2}{R} \therefore R = \frac{52900}{920} = 57.5\Omega$																																			
5	B	☒ A There is no change in Temperature so there is no change in Kinetic Energy ☒ B Increase in pressure means more air particles hit the sides of the syringe ☒ C There are more collisions with the sides of the syringe when pressure is increased ☒ D There are less collisions with the sides of the syringe when pressure is decreased ☒ E There is no change in Temperature so there is no change in Kinetic Energy																																			
6	D	Volume: Constant $p_1 = 150 \text{ kPa}$ $T_1 = 27^\circ\text{C} = 300 \text{ K}$ $p_2 = ?$ $T_2 = 47^\circ\text{C} = 320 \text{ K}$ $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ $\frac{150}{300} = \frac{p_2}{320}$ $p_2 = \frac{150 \times 320}{300}$ $p_2 = 160 \text{ kPa}$																																			
7	D	3 complete waves = 3 wavelengths = 18mm = 1 wavelength = 6mm																																			
8	A	Statement I - Correct		Statement II - Incorrect		Statement III - Incorrect																															
		Light waves are transverse waves. All forms of electromagnetic radiation are transverse waves		Radio waves travel at $3 \times 10^8 \text{ m s}^{-1}$ like all other forms of electromagnetic radiation. Sound waves travel at 340 m s^{-1}		<table><tr><td>EM Type</td><td>Gamma</td><td>X-Ray</td><td>Ultra-violet</td><td>Visible</td><td>Infra-Red</td><td>Microwave</td><td>Radio & TV</td></tr><tr><td>Energy</td><td>High</td><td colspan="4">←————→</td><td>Low</td></tr><tr><td>Frequency</td><td>High</td><td colspan="4">←————→</td><td>Low</td></tr><tr><td>Wavelength</td><td>Low</td><td colspan="4">←————→</td><td>High</td></tr></table>			EM Type	Gamma	X-Ray	Ultra-violet	Visible	Infra-Red	Microwave	Radio & TV	Energy	High	←————→				Low	Frequency	High	←————→				Low	Wavelength	Low	←————→				High
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Ionising describes the process where atoms <i>lose</i> electrons and become <i>positively</i> charged or atoms <i>gain</i> electrons and become <i>negatively</i> charged																																					
9	C																																				

10	E	<table><tr><th>Statement I - Correct</th><th>Statement II - Incorrect</th><th>Statement III - Correct</th></tr><tr><td>Shielding of tissue with a lead screen will reduce the equivalent dose received</td><td>Increasing the distance from the source will <u>reduce</u> the equivalent dose received by the tissue</td><td>Increasing the time of exposure will increase the equivalent dose received by the tissue</td></tr></table>	Statement I - Correct	Statement II - Incorrect	Statement III - Correct	Shielding of tissue with a lead screen will reduce the equivalent dose received	Increasing the distance from the source will <u>reduce</u> the equivalent dose received by the tissue	Increasing the time of exposure will increase the equivalent dose received by the tissue												
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11	E	<table><tr><td>H = ?</td><td>D = 16μGy = 16x10⁻⁶ Gy</td><td>w_R = 20</td></tr><tr><td></td><td>H = D x w_R</td><td></td></tr><tr><td></td><td>H = 16x10⁻⁶ x 20</td><td></td></tr><tr><td></td><td>H = 320x10⁻⁶ Sv</td><td></td></tr><tr><td></td><td>= 320 μSv</td><td></td></tr></table>	H = ?	D = 16μGy = 16x10 ⁻⁶ Gy	w _R = 20		H = D x w _R			H = 16x10 ⁻⁶ x 20			H = 320x10 ⁻⁶ Sv			= 320 μSv				
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12	A	<table><tr><td>A = ?</td><td>N = 240</td><td>t = 1 minute = 60s</td></tr><tr><td colspan="3">$A = \frac{N}{t} = \frac{240}{60} = 4 \text{ Bq}$</td></tr></table>	A = ?	N = 240	t = 1 minute = 60s	$A = \frac{N}{t} = \frac{240}{60} = 4 \text{ Bq}$														
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13	E	<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>Fusion Nuclear fusion occurs when two smaller nuclei join together to make a bigger nuclei</td><td>Fission Nuclear fission occurs when a larger nuclei splits into two smaller nuclei</td><td>Energy The release of energy in fission and fusion reactions can be used as an energy source.</td></tr></table>	X	Y	Z	Fusion Nuclear fusion occurs when two smaller nuclei join together to make a bigger nuclei	Fission Nuclear fission occurs when a larger nuclei splits into two smaller nuclei	Energy The release of energy in fission and fusion reactions can be used as an energy source.												
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14	C	<table><tr><td colspan="6">Vector quantities have magnitude <u>and</u> direction while scalar quantities only have magnitude.</td></tr><tr><td>Vector Quantity</td><td>force</td><td>velocity</td><td>displacement</td><td>acceleration</td><td>weight</td></tr><tr><td>Scalar Quantity</td><td>energy</td><td>speed</td><td>distance</td><td>time</td><td>mass</td></tr></table>	Vector quantities have magnitude <u>and</u> direction while scalar quantities only have magnitude.						Vector Quantity	force	velocity	displacement	acceleration	weight	Scalar Quantity	energy	speed	distance	time	mass
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15	B	<table><tr><th>Statement I - Incorrect</th><th>Statement II - Correct</th><th>Statement III - Incorrect</th></tr><tr><td>X is accelerating as its velocity is increasing by 2 m s⁻¹ every second</td><td>Y is undergoing constant acceleration as velocity is increasing by 1 m s⁻¹ every second</td><td>Z is <i>not</i> undergoing constant acceleration as the velocity is not increasing by same amount every second.</td></tr></table>	Statement I - Incorrect	Statement II - Correct	Statement III - Incorrect	X is accelerating as its velocity is increasing by 2 m s ⁻¹ every second	Y is undergoing constant acceleration as velocity is increasing by 1 m s ⁻¹ every second	Z is <i>not</i> undergoing constant acceleration as the velocity is not increasing by same amount every second.												
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16	C	<table><tr><td>E_w = ?</td><td>F = 2500N</td><td>d = 50m</td></tr><tr><td></td><td>E_w = F x d</td><td></td></tr><tr><td></td><td>E_w = 2500 x 50</td><td></td></tr><tr><td></td><td>E_w = 125000 J</td><td></td></tr></table>	E _w = ?	F = 2500N	d = 50m		E _w = F x d			E _w = 2500 x 50			E _w = 125000 J							
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	E _w = F x d																			
	E _w = 2500 x 50																			
	E _w = 125000 J																			
17	A	☒A The reaction force is the chair on the person ☒B This is the same force not the reaction force ☒C The person is not touching the Earth directly so there is no reaction force of the Earth on the person ☒D The force of the chair on Earth is not the reaction force to the person sitting on the chair ☒E The person is not directly applying force to Earth as the chair is applying the force on Earth																		
18	B	<table><tr><th>Statement I - Incorrect</th><th>Statement II - Correct</th><th>Statement III - Incorrect</th></tr><tr><td>At terminal velocity the weight of the balance is equal to the air resistance acting on the passage</td><td>As terminal velocity, the object is travelling at constant velocity indicating forces are balanced</td><td>The object is travelling at a constant velocity at terminal velocity so not accelerating</td></tr></table>	Statement I - Incorrect	Statement II - Correct	Statement III - Incorrect	At terminal velocity the weight of the balance is equal to the air resistance acting on the passage	As terminal velocity, the object is travelling at constant velocity indicating forces are balanced	The object is travelling at a constant velocity at terminal velocity so not accelerating												
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19	E	<table><tr><td>d = v x t</td></tr><tr><td>d = 3x10⁸ x 4.3 x 365.25 x 24 x 60 x 60</td></tr><tr><td>d = 4.1x10¹⁶ m</td></tr></table>	d = v x t	d = 3x10 ⁸ x 4.3 x 365.25 x 24 x 60 x 60	d = 4.1x10 ¹⁶ m															
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20	D	<table><tr><td>Line Spectrum From Star</td><td></td><td></td></tr><tr><td>Calcium</td><td></td><td></td></tr><tr><td>Helium</td><td></td><td></td></tr><tr><td>Hydrogen</td><td></td><td></td></tr><tr><td>Sodium</td><td></td><td></td></tr></table>	Line Spectrum From Star			Calcium			Helium			Hydrogen			Sodium					
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Question	Answer	Physics Covered											
1a		<table><tr><th>Cell</th><th>Lamp</th><th>Resistor</th><th>Switch</th></tr><tr><td></td><td></td><td></td><td></td></tr></table>	Cell	Lamp	Resistor	Switch							
Cell	Lamp	Resistor	Switch										
													
1b	5 Ω	$V = 2.5 \text{ V}$ $I = 0.5 \text{ A}$ $R = ?$ $V = I R$ (1 mark) $2.5 = 0.5 \times R$ (1 mark) $R = 5 \Omega$ (1 mark)											
1c	Answer to include:	<table><tr><td>Effect (1 mark) (1 mark) (lamp L/it is) brighter</td><td colspan="3">Justification (2 marks) (1 mark) M is in parallel (with resistor) Greater current in/through lamp L (than that in M) or Greater voltage across lamp L (than across M)</td></tr></table>	Effect (1 mark) (1 mark) (lamp L/it is) brighter	Justification (2 marks) (1 mark) M is in parallel (with resistor) Greater current in/through lamp L (than that in M) or Greater voltage across lamp L (than across M)									
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2a	<u>1 mark</u> : Graph X <u>1 mark</u> : LED only conducts in one direction	LED are light emitting diodes only conduct electricity in on direction in a circuit. Graphs Y & Z can be ruled out as they show that electricity flowing in both directions. Graph X has a positive voltage & current but has no negative voltage & current indicating current flowing in one direction only.											
2b(i)	120 J	$P = ?$ $V = 4.0 \text{ V}$ $I = 0.50 \text{ A}$ $P = \frac{V}{I}$ (1 mark) $P = 4.0 \times 0.50$ $P = 2.0 \text{ W}$ $P = 2.0 \text{ W}$ $E = ?$ $t = 60 \text{ s}$ $P = \frac{E}{t} \therefore 2.0 = \frac{E}{60} \therefore E = 2.0 \times 60 = 120 \text{ J}$ (1 mark) (1 mark) (1 mark)											
2b(ii)	30 C	$Q = ?$ $I = 0.50 \text{ A}$ $t = 60 \text{ s}$ $Q = \frac{I}{t}$ (1 mark) $Q = 0.5 \times 60$ $Q = 30 \text{ C}$											
3a(i)	15ms	Position Y is thinner than position X. Position Y will reflect pulse first at 5ms and position X will reflect pulse second at 15ms											
3a(ii)	0.039m	$d = ?$ $v = 5200 \text{ m s}^{-1}$ $t = 15 \text{ ms} = 15 \times 10^{-6} \text{ s}$ $d = \frac{v}{t} \times t$ (1 mark) $d = 5200 \times 15 \times 10^{-6}$ (1 mark) $d = 0.0078 \text{ m}$ (1 mark) Thickness = $\frac{0.078}{2} = 0.039 \text{ m}$ (1 mark)											
3b		<table><tr><td>1 mark</td><td>Vertical line between 5 and 15 on x-axis</td></tr><tr><td>1 mark</td><td>Amplitude between 25 and 40 on y-axis</td></tr></table> The imperfection at Position Z is between the thickness at Positions X and Y meaning the time taken and amplitude of the reflected pulse will be between the values at Position X and Y				1 mark	Vertical line between 5 and 15 on x-axis	1 mark	Amplitude between 25 and 40 on y-axis				
1 mark	Vertical line between 5 and 15 on x-axis												
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3c(i)	2.5x10 ⁵ Hz or 250,000Hz	$f = ?$ $T = 4.0 \mu\text{s} = 4 \times 10^{-6} \text{ s}$ $f = \frac{1}{T}$ (1 mark) $f = \frac{1}{4 \times 10^{-6}}$ (1 mark) $f = 2.5 \times 10^5 \text{ Hz}$											

3c(ii)	0.021 m	$v = 5200 \text{ m s}^{-1}$ $f = 2.5 \times 10^{-5} \text{ Hz}$ $\lambda = ?$ $v = f \times \lambda$ (1 mark) $5200 = 2.5 \times 10^{-5} \times \lambda$ (1 mark) $\lambda = 0.021 \text{ m}$ (1 mark)																															
3d	Answer to include:	1 mark	Speed of ultrasound in brass is less than in steel																														
		1 mark	Takes greater time to travel same distance/thickness.																														
4	Open Ended Question	1 mark	2 marks	3 marks																													
		Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.																													
5a		Angle of incidence is measured from the normal to the light ray going into the diamond Angle of refraction is measured from the normal to the light ray inside the diamond																															
5b	Decreases	When light passes into a diamond the speed of light decreases																															
5c	B	The higher the optical density the higher the degree of refraction. This leads to a greater bending of light towards to the normal so the angle of refraction would be the smallest with the greater degree of refraction																															
5d	$5.5 \times 10^9 \text{ Pa}$	$P = ?$	$F = 61 \text{ kN} = 61000 \text{ N}$ $P = \frac{F}{A} = \frac{61000}{1.1 \times 10^{-5}} = 5.5 \times 10^9 \text{ Pa}$ (1 mark)	$A = 1.1 \times 10^{-5} \text{ m}^2$ (1 mark)																													
6a	increases	The thicker the paper the lower the quantity of radiation that can cross the paper. By making the paper thinner then the quantity of radiation crossing the paper will increase.																															
6b(i)	Answer to include:	Choice of Radioactive Source	Explanation																														
		X	Beta Source Required	Long half-life																													
		(1 mark)	(1 mark)	(1 mark)																													
6b(ii)	One answer from:	Time for activity to (decrease by) half Time for half the nuclei to decay																															
6b(iii)	(high frequency) electromagnetic wave	Gamma radiation is one form of electromagnetic radiation: <table border="1"><thead><tr><th>EM Type</th><th>Gamma</th><th>X-Ray</th><th>Ultra-violet</th><th>Visible</th><th>Infra-Red</th><th>Microwave</th><th>Radio & TV</th></tr></thead><tbody><tr><td>Energy</td><td>High</td><td colspan="4">←————→</td><td>Low</td></tr><tr><td>Frequency</td><td>High</td><td colspan="4">←————→</td><td>Low</td></tr><tr><td>Wavelength</td><td>Low</td><td colspan="4">←————→</td><td>High</td></tr></tbody></table>			EM Type	Gamma	X-Ray	Ultra-violet	Visible	Infra-Red	Microwave	Radio & TV	Energy	High	←————→				Low	Frequency	High	←————→				Low	Wavelength	Low	←————→				High
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6c	2 hours	 Count rate halves from 600kBq to 300kBq Time Interval = 3.4 hours – 1.4 hours = 2 hours																															
		Take any halving of the corrected count rate on the y-axis. Work out the time interval on the x-axis for this halving.																															

7a(i)	$10 \times 10^3 \text{ N}$ or $1.0 \times 10^4 \text{ N}$ or 10000 N	$8.0 \times 10^3 \text{ N}$ θ $6.0 \times 10^3 \text{ N}$ x $x = \sqrt{(6000)^2 + (8000)^2}$$x = \sqrt{36000000 + 64000000}$$x = \sqrt{100000000}$$x = 10000 \text{ km} = 10 \times 10^3 \text{ km} = 1 \times 10^4 \text{ km}$						
7a(ii)	37°	$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{6.0 \times 10^3}{8.0 \times 10^3} = 0.75 \quad \therefore \theta = 37^\circ$						
7a(iii)	$2 \times 10^{-3} \text{ m s}^{-2}$	$F = 10 \times 10^3 \text{ N}$ $m = 5.0 \times 10^6 \text{ kg}$ $a = ?$ $F = m a$ (1 mark) $10 \times 10^3 = 5.0 \times 10^6 \times a$ (1 mark) $a = 2.0 \times 10^{-3} \text{ m s}^{-2}$ (1 mark)						
7b	Answer to include:	<u>1 mark</u> 1 from: Buoyancy force Upthrust force of water on ship flotation force  <u>1 mark</u> 1 from: Weight Force of Gravity <u>1 mark</u> (These) forces are balanced						
8a(i)	Answer to include:	<table><tr><td><u>1 mark</u></td><td><u>1 mark</u></td><td><u>1 mark</u></td></tr><tr><td>length/width of card</td><td>time taken for card to pass (through) the light gate</td><td>time taken (for trolley to travel from starting position) to light gate</td></tr></table>	<u>1 mark</u>	<u>1 mark</u>	<u>1 mark</u>	length/width of card	time taken for card to pass (through) the light gate	time taken (for trolley to travel from starting position) to light gate
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8a(ii)	One answer from:	<table><tr><td>reaction time (can cause error with the stop clock reading)</td><td>card may not have passed straight through light gate</td><td>length/width of card not measured properly (e.g. ruler not straight along card)</td><td>other suitable reason</td></tr></table>	reaction time (can cause error with the stop clock reading)	card may not have passed straight through light gate	length/width of card not measured properly (e.g. ruler not straight along card)	other suitable reason		
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8b	0.64 m s^{-2}	$a = ?$ $v = 1.6 \text{ m s}^{-1}$ $u = 0 \text{ m s}^{-1}$ $t = 2.5 \text{ s}$ $a = \frac{v - u}{t} = \frac{1.6 - 0}{2.5} = 0.64 \text{ m s}^{-2}$ (1 mark) (1 mark) (1 mark)						
9a		A suitable curved path where the ball does not increase in height. The stone will fall vertically faster the further it falls due to gravity The horizontal velocity will remain the same						
9b(i)	7.8 m s^{-1}	$a = 9.8 \text{ m s}^{-2}$ $v = ?$ $u = 0 \text{ m s}^{-1}$ $t = 0.80 \text{ s}$ $a = \frac{v - u}{t}$ (1 mark) $9.8 = \frac{v - 0}{0.80}$ (1 mark) $9.8 \times 0.80 = v - 0$ $7.8 \text{ m s}^{-1} = v$ (1 mark)						

9b(ii)	3.1 m	$\bar{v} = ?$ $\bar{v} = \frac{u + v}{2} = \frac{0 + 7.8}{2} = \frac{7.8}{2} = 3.9 \text{ m s}^{-1} \quad (1 \text{ mark})$ $v = 7.8 \text{ m s}^{-1}$ $d = ?$ $d = \bar{v} \times t \quad (1 \text{ mark})$ $t = 0.80 \text{ s}$ $d = 3.9 \times 0.80 \quad (1 \text{ mark})$ $d = 3.1 \text{ m} \quad (1 \text{ mark})$										
9c	same time	The time taken for a stone to hit the water when dropped vertically is the same as the time taken when thrown horizontally as the vertical component is subject to the same acceleration due to the gravitational field strength being the same.										
10	Open Ended Question	<table><tr><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</td><td>Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</td><td>Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</td></tr></table>	1 mark	2 marks	3 marks	Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.				
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11a(i)	0.20 J	$E_p = ? \quad m = 0.040 \text{ kg} \quad g = 9.8 \text{ N kg}^{-1} \quad h = 0.50 \text{ m}$ $E_p = m \times g \times h \quad (1 \text{ mark})$ $E_p = 0.040 \times 9.8 \times 0.50 \quad (1 \text{ mark})$ $E_p = 0.20 \text{ J} \quad (1 \text{ mark})$										
11a(ii)	One answer from:	<table><tr><td>kinetic energy to heat (and sound)</td><td>kinetic energy of marble to kinetic energy of sand.</td></tr></table>	kinetic energy to heat (and sound)	kinetic energy of marble to kinetic energy of sand.								
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11b(i)	Graph showing:	<table><tr><td>1 mark</td><td>1 mark</td><td>1 mark</td></tr><tr><td>suitable scales, labels and units</td><td>all points plotted accurately to $\pm$ half a division</td><td>best fit curve</td></tr></table>	1 mark	1 mark	1 mark	suitable scales, labels and units	all points plotted accurately to $\pm$ half a division	best fit curve				
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11b(ii)												
11b(iii)	Any two from:	<table><tr><td colspan="5">1 mark for each answer:</td></tr><tr><td>Repeat (and average)</td><td>Take (more) readings in the 0.15m to 0.35m drop height range</td><td>Increase the height range</td><td>level sand between drops</td><td>or other suitable improvement</td></tr></table>	1 mark for each answer:					Repeat (and average)	Take (more) readings in the 0.15m to 0.35m drop height range	Increase the height range	level sand between drops	or other suitable improvement
1 mark for each answer:												
Repeat (and average)	Take (more) readings in the 0.15m to 0.35m drop height range	Increase the height range	level sand between drops	or other suitable improvement								
11c(i)	One variable from:	<table><tr><td>mass/weight of marble</td><td>angle of impact</td><td>diameter of marble</td><td>type of sand</td></tr><tr><td>radius of marble</td><td>density of marble</td><td>volume of marble</td><td>speed of marble</td><td>time of drop</td></tr></table>	mass/weight of marble	angle of impact	diameter of marble	type of sand	radius of marble	density of marble	volume of marble	speed of marble	time of drop	
mass/weight of marble	angle of impact	diameter of marble	type of sand									
radius of marble	density of marble	volume of marble	speed of marble	time of drop								
11c(ii)	Answer to include:	<table><tr><td>1 mark</td><td>How independent variable can be measured/changed</td></tr><tr><td>1 mark</td><td>State at least one other variable to be controlled</td></tr></table>	1 mark	How independent variable can be measured/changed	1 mark	State at least one other variable to be controlled						
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